MIL-C-2070E
31 December 1985
SUPERSEDING
MIL-C-2070D
15 December 1976

MILITARY SPECIFICATION

CHAIN ASSEMBLIES, SINGLE LEG, FOR CABINET, GASOLINE, FIELD RANGE OUTFIT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a right and left chain assembly for the field range cabinet (see 6.1).

1.2 Classification. Chain assemblies shall be of the following styles, as specified (see 6.2).

Style 1 - Right-hand
Style 2 - Left-hand

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-T-791	- Turnbuckle
QQ-W-461	- Wire, Steel, Carbon, Round, Bare, and Coated

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 4010

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RR-C-271 - Chains and Attachments, Welded and Weldless
PPP-B-601 - Boxes, Wood, Cleated-Plywood

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

5-11-90 - Chains, Single Leg, for Cabinet, Gasoline, Field Range Outfit

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

* 3.1 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer for the chain assemblies, provided the finished item complies with all provisions of this document. Recycled material may be used (see 6.3).

3.1.1 Wire, carbon steel. Carbon steel wire shall conform to steel number 1006 or 1010, finish 1 of QQ-W-461.

3.1.2 Chain. Chain shall be nominal 3/16-inch, type I, grade C, class 4, conforming to RR-C-271.

3.1.3 Turnbuckles. Turnbuckles shall be nominal 3/8-inch, type I, form 1, class 4 (right-hand) or class 6 (left-hand), zinc coated, conforming to FF-T-791 except threads shall be a class 2 fit after zinc coating.

3.2 Construction. Chain assemblies shall be constructed as specified herein and as shown on Drawing 5-11-90. The completed assemblies shall be capable of withstanding a pull of 520 pounds, at a rate not to exceed 600 pounds per minute without breaking or deformation when tested as specified in 4.3.2.

3.2.1 Hook and half-link. The hook shall be formed of the steel specified in 3.1.1. The half-link shall be welded to the hook or the hook and half-link shall be of drop-forged construction.

3.2.2 Turnbuckle. The threaded end of the eye-pulls and hook shall be staked or flattened as shown on the drawing. The eye-pull on the right-hand turnbuckle shall be offset 23 degrees, plus or minus 2 degrees.

* 3.3 Finish. After complete fabrication, and prior to zinc coating, components shall have all rough and sharp edges removed and shall be thoroughly cleaned of all grease, oil, dirt, and foreign matter. Components not zinc coated shall be given either a minimum 0.0005-inch thick hot-dip zinc coating in accordance with standard commercial practice or be electro-zinc-plated in accordance with the requirements for type I, classification Fe/Zn13, Sc3 of ASTM B 633 (see 4.3.1). All dipped or plated items shall be smooth, continuously coated, and free of lumps and drippings, and shall not be flaked, peeled, or pitted.

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3.4 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be continuous, sound, smooth, and free of cracks, burn holes, undercuts, pits, fissures, or incomplete fusion. All scale shall be removed from the finish weld area.

3.5 Fastening devices. Threaded fasteners shall not be broken, cracked, or stripped.

3.6 Workmanship. All components and assemblies of the unit shall be free from dirt and other extraneous material, burrs, slivers, rough die, tool and grind marks, dents, and cracks. Castings and molded parts, if used, shall be free of sand, fins, pits, blow holes, and sprues. External surfaces shall be free from sharp edges and sharp corners. Hook end shall not be rounded.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 In-process inspection. After fabrication and prior to zinc coating, the components shall be examined to determine compliance with the cleaning requirements of 3.3. Whenever nonconformance is noted, correction shall be made to affected items and to the operation and process.

4.2.3 End item inspection. The lot shall consist of all chain assemblies of one style offered for inspection at one time. The sample unit shall be a complete chain assembly.

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* 4.2.3.1 Visual examination. Chain assemblies shall be examined for defects listed in table I. The inspection level shall be II, and the acceptable quality level (AQL) shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not hot-dip or electroplated as specified	101	
	Not smooth and continuous coating	102	
	Any lumps or drips		201
	Any evidence of flaking, peeling, or pitting	103	
Construction and workmanship (general)	Not style specified	104	
	Any burr, sliver, sharp edge, rough die, tool, gouge, trim, or grind marks	105	
	Any component missing or improperly located	106	
	Any component cracked or malformed	107	
Welding	Missing	108	
	Not continuous, not smooth, not uniform, burn holes, cracks, fissures, incomplete fusion	109	
	Not finished to blend to adjacent surface	110	
	Not free from flux, scale, paint, or areas of other foreign material	111	

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TABLE I. Classification of defects - Continued

Examine	Defect	Classification	
		Major	Minor
Threaded fasteners	Missing, stripped or cross-threaded, fractured	112	
	Loose		202
	Threaded ends of eye-pulls and hook not flattened or staked	113	
	Hook and half-link not welded or drop forged	114	

4.2.3.2 Dimensional examination. Examination shall be made of the finished chain assemblies to determine conformance with dimensional requirements. Any nonconformance to specified requirements shall be classified a defect. The inspection level shall be S-2, and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.2.3.3 End item testing. Chain assemblies shall be tested as specified in 4.3.1 and 4.3.2 using inspection level S-2, and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.2.4 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the requirements of section 5. Defects shall be as indicated in table II. The lot shall consist of the number of containers offered for inspection at one time. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the AQL shall be 4.0, expressed in terms of defects per hundred units.

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TABLE II. Packaging inspection

Examine	Defect
Markings, exterior and interior	Omitted, incorrect, illegible, or of improper size, location, sequence, or method of application
Materials	Component missing, damaged, or otherwise defective
Workmanship	Item not coiled or secured as specified Bulged or distorted containers
Contents	Number not as specified

4.3 Tests.

* 4.3.1 Thickness of zinc coating. The zinc coating thickness shall be tested in accordance with standard commercial practice when the chain assemblies are hot-dipped and as specified in ASTM B 633 when electro-zinc plated to determine conformance to 3.3. Inability to meet the coating thickness requirement of 3.3 shall constitute failure of this test.

4.3.2 Pull test. The hook shall be fastened at one end of a tension testing machine and the turnbuckle eye or hook, as applicable, shall be fastened at the other end. The chain shall then be pulled with a uniform application of load at a rate not to exceed 600 pounds per minute until a load of 520 pounds has been reached. The chain assembly shall then be removed from the machine and checked to determine conformance to 3.2. Inability to meet the strength requirements of 3.2 shall constitute failure of this test.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

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5.1.1 Level A. Each chain assembly shall be coiled to its smallest practicable dimension and securely wired in a minimum of two places with annealed wire of not less than 0.0720-inch diameter.

* 5.1.2 Commercial. Chain assemblies shall be preserved in accordance with ASTM D 3951.

* 5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Thirty chain assemblies, preserved as specified in 5.1, shall be packed in a snug-fitting, cleated-plywood shipping container conforming to overseas type, style A or J, grade A or B, type 3 load of PPP-B-601. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-601.

5.2.2 Level B packing. Thirty chain assemblies, preserved as specified in 5.1, shall be packed in a snug-fitting, cleated-plywood shipping container conforming to domestic type, grade B, type 3 load of PPP-B-601.

* 5.2.3 Commercial packing. Chain assemblies, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable. Each chain assembly shall have a tag attached for the identification information.

6. NOTES

6.1 Intended use. Chain assemblies are used to secure the Cabinet, Gasoline, Field Range Outfit - MIL-C-40077, to the body of field kitchen vehicles while in transit.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.

b. Style required (see 1.2).

c. Selection of applicable levels of preservation and packaging (see 5.1 and 5.2).

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* 6.3 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.1).

6.4 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-2070E	Chain Assemblies, Single Leg, For Cabinet, Gasoline, Field Range Outfit

NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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PROBLEM AREAS

a. Paragraph Number and Wording:

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6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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